Deepak Kharol

Senior Software Engineer, CleverTap
Mumbai, Maharashtra, India
✉ contact@deepakkharol.com in linkedin.com/in/deepakharol

Summary

Backend-focused Software Engineer with a strong track record of building scalable, data-intensive systems used in high-throughput production environments. Experienced in owning complex backend features end-to-end—from system design and data modeling to performance optimization and operational reliability. Known for solving growth-driven problems, leading critical initiatives, and mentoring engineers while delivering measurable business impact.

Experience

- **CleverTap** — Mumbai, Maharashtra, India
  - *Senior Software Engineer* Jan 2024 – Present
    - **Cross-Channel Message Frequency Capping:** Architected a rule-based platform enforcing per-category, per-channel, and cross-channel limits on messages sent to a user, preventing over-messaging at scale. Designed for extensibility—became the foundation for **5 downstream features** (segment-level caps, campaign prioritization, delivery simulation, message-spacing controls, regulatory frequency compliance), significantly reducing their build time.
    - **Delivery Simulation & Message Spacing:** Built a simulator predicting which campaign messages a user would receive vs. get blocked under configured limits—adopted by large enterprise clients including **Nykaa, Bajaj, and Axis Bank**; delivered configurable minimum time-gap enforcement between consecutive messages to a user.
    - **PII Encryption:** Led platform-wide encryption of sensitive user data (profile attributes, email, phone) with controlled runtime decryption—strengthening the platform's data security and compliance posture without impacting campaign delivery performance.
    - **Data Archival Pipeline:** Built an automated pipeline offloading inactive campaign targeting data from the primary MongoDB cluster to a dedicated archival datastore, improving primary cluster performance and stability; owned rollout and production hardening.
    - **Server-Side App Inbox:** Designed a Redis-backed inbox system for storing and delivering in-app user messages, built for **StockX**; contributed to migration of **$1M ARR**.
    - **Dynamic Content Personalization:** Architected external API–driven message personalization, fetching content from client systems at send time to generate dynamic payloads and reduce client onboarding friction; also shipped multi-version support for campaign message content.
    - **Leadership & Reliability:** POC for a 10-member team; mentored 3 engineers delivering 5 production features in 2 months; drove cross-team incident resolution as part of on-call ownership; **won the company-wide Hackathon**.
  - *Software Engineer* Jul 2022 – Dec 2023
    - **Campaign API Platform:** Developed APIs enabling clients to create Push, Email, and SMS campaigns programmatically; currently powering **40%+** of all campaigns on the platform.
    - **Audience & Ads Integration:** Integrated Facebook and Google Ads workflows for audience creation and ad set targeting using profile attributes such as email and phone number.
    - **Push Notification Analytics:** Built analytics pipelines and dashboards to compute push notification render rates, improving campaign performance visibility.
  - *Backend Intern* Dec 2021 – Jun 2022
    Contributed to backend services, production fixes, and performance improvements.
- **Dell Technologies** — Bengaluru, India May 2021 – Jul 2021
  Business Intelligence Intern — Built dashboards and delivered insights using Power BI and Tableau.

Education

- **Kalinga Institute of Industrial Technology, Bhubaneswar**
  B.Tech in Computer Science **(CGPA: 9.36)** Jul 2018 – Jun 2022
- **AECS Rawatbhata, Rajasthan** — Class XII (CBSE) **91.4%** (2017); Class X (CBSE) **9.4 CGPA** (2015)

Skills

- **Languages & Systems:** Java, C++, Python, Redis, MongoDB, RocksDB
- **Backend Engineering:** Distributed Systems, API Design, Data Modeling, Performance Optimization
- **Professional Strengths:** Technical Leadership, Structured Problem Solving, Mentorship

Awards & Interests

- Winner, CleverTap company-wide Hackathon (2025); KIIT Merit Scholarship; Microsoft MTA Certification (Python)
- Interests: Long-distance bike riding, competitive football, chess (FIDE-rated tournament participant), hiking